Wenshuo Wang

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EDUCATION

South China University of Technology, Guangzhou, China **Sep 2023 - Present**

B.Eng, School of Future Technology, Major in Artificial Intelligence

- **GPA:** 3.6/4.0
- **Major Courses:** Fluid Mechanics, Analysis and Applications of Heat, Engineering Math: Calculus II, Engineering Math:Probability & Mathematical Statistics, Discrete Mathematics, Python, Data Structures, Machine Learning, Deep Learning and Computer Vision, Introduction to Cognitive Neuroscience, and related subjects
- **Languages:** English (TOEFL iBT 101)

Massachusetts Institute of Technology, Cambridge, MA, USA **Nov 2024 - Present**

Research Intern (Remote), Kavli Institute for Astrophysics and Space Research (Advisor: Prof. Fan Zhang)

RESEARCH INTERESTS

My research focuses on **deep learning for scientific computing**, especially **neural PDE simulation** for physical-field dynamics. I study neural PDE simulators beyond architecture design, focusing on how problem states, observations, representations, and computational budgets determine rollout fidelity and physical reliability. My long-term goal is to develop efficient, reliable, and physically faithful neural simulation methods for scientific and engineering systems.

PUBLICATIONS AND PREPRINTS

[1] Posterior-First Neural PDE Simulation: Inferring Hidden Problem State from a Single Field

Preprint [**Paper**]

Wenshuo Wang, *Fan Zhang*† *(*† *Corresponding Author)*

- Theoretically establish that a single observed field can leave multiple task-relevant problem states unresolved, making posterior problem-state inference necessary before reliable neural PDE rollout, rather than deterministic field-to-future prediction.
- Introduce a posterior-first simulator that improves rollout fidelity and hidden-physics recovery.

[2] Derived Fields Preserve Fine-Scale Detail in Budgeted Neural Simulators

Preprint [**Paper**]

Wenshuo Wang, *Fan Zhang*†

- Reposition carried-state design as the primary bottleneck in budgeted neural simulation by showing that, under tight storage budgets, fine-scale fidelity is often determined when the state is constructed, before stronger architectures, losses, or rollout strategies can recover information already discarded.
- Propose DerivOpt, a budgeted state-design framework that optimizes which physical fields a neural simulator carries and how storage is allocated across them.

[3] Breaking Scale Anchoring: Frequency Representation Learning for Accurate High-Resolution Inference from Low-Resolution Training

ICLR 2026 [**Paper**]

Wenshuo Wang, *Fan Zhang*†

- Identify Scale Anchoring and propose Frequency Representation Learning for more accurate high-resolution spatiotemporal inference from low-resolution training.
- Correct the field's notion of multi-resolution generalization by showing that stable error across resolutions can mask a deeper failure to learn the true physical operator.

[4] Gradient-Informed Temporal Sampling Improves Rollout Accuracy in PDE Surrogate Training

Preprint [**Paper**]

Wenshuo Wang, *Fan Zhang*†

- Challenge the field's default reliance on systematic temporal sampling by showing that neural-simulator training windows should be selected rather than uniformly retained, and that existing generic samplers are not well aligned with this task.
- Propose GITS, a temporal sampling method that combines pilot-model gradient scores with set-level temporal coverage for PDE surrogate training.

[5] System-Anchored Knee Estimation for Low-Cost Context Window Selection in PDE Forecasting

Preprint [**Paper**]

Wenshuo Wang, *Fan Zhang*†

- Shift context-window selection from routine hyperparameter tuning to low-cost algorithmic estimation of near-optimal windows, avoiding exhaustive sweeps for neural PDE simulators.
- Propose SAKE, a two-stage low-cost method that uses system-derived anchors and knee-aware search to select context windows for autoregressive neural PDE simulators.

[6] S^2-KD: Semantic-Spectral Knowledge Distillation Spatiotemporal Forecasting

AAAI 2026 [**Paper**]

Wenshuo Wang, *Yaomin Shen, Yingjie Tan, Yihao Chen*†

- Recast spatiotemporal distillation as transferring semantic and causal understanding, not just pixel- or frequency-level patterns.
- Propose S^2-KD, a semantic-spectral distillation framework that transfers causal semantics and spectral knowledge from a multimodal teacher to a vision-only student for spatiotemporal forecasting.

[7] iTAG: Inverse Design for Text Generation with Accurate Causal Graph Annotations

ACL 2026 (Oral Presentation) [**Paper**]

Wenshuo Wang, *Boyu Cao, Nan Zhuang, Wei Li*†

- Propose iTAG, an inverse-design framework that assigns real-world concepts to causal graphs and generates natural text with highly accurate causal annotations.
- Enable low-cost, usable surrogate data for text-based causal discovery when real-world causally annotated text is scarce and expensive to obtain.

[8] Pi-CCA: Prompt-Invariant CCA Certificates for Replay-Free Vision–Language Continual Learning

ICLR 2026 [**Paper**]

Jiayu Zhang, Chuangxin Zhao, Ruibo Duan, Canran Xiao, Wenyi Mo, Haoyu Gao, ***Wenshuo Wang***†

- Reframe replay-free vision–language continual learning around preserving alignment invariants under prompt and domain shifts, rather than matching proxy outputs.
- Introduce PI-CCA, a replay-free continual multimodal learning method that preserves image–text alignment geometry with compact CCA certificates.

INTERNSHIP EXPERIENCE

HUAWEI Beijing Research Institute, Beijing, China **Jul 2024 - Oct 2024**

Algorithm Engineer Intern

- Developed NLP and medical AI applications for text understanding, summarization, retrieval, and conversational question answering.
- Conducted research on NLP, focusing on LLMs.